

MATH 1342 – Elementary Statistics – Week 14: Chi-Square Test for Association

Nov 23 – 25 | 7.2 Testing for an Association Between Two Categorical Variables • No Class – College closes at noon (Wed Nov 25) | Fall 2026 • Section 019 • MW 7:30–8:50 PM • RRC Room 8210

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
4. Find and use theoretical probabilities from normal, t, chi-squared and F distributions to form confidence intervals and test hypotheses. Apply the 95% rule to normal and to approximately normal distributions. 6. Analyze data presented in two-way tables to provide information about relationships between categorical variables. 11. Demonstrate skills in hypothesis testing using chi-squared tests.	1. State the hypotheses for a chi-square test of association from a two-way table. (CO6, CO11) 2. Compute expected counts and check that all expected counts are at least 5. (CO11) 3. Compute the chi-square statistic and $df = (r - 1)(c - 1)$, and interpret the right-tail p-value. (CO4, CO11) 4. Relate the chi-square test on a 2x2 table to the two-proportion z test, and explain that a significant result shows association but not direction or causation. (CO11)	e-Text: 7.2 Testing for an Association Between Two Categorical Variables	All	Participation: Chi-Square Test – faculty rank × gender: compute expected counts, cell contributions, χ^2 and df; decide and interpret	MO1, MO2, MO3, MO4	Online Homework: 7.2 Testing for an Association Between Two Categorical Variables (due Sun Nov 29, 11:59 PM)	All
		Interactive Lecture: 7.2 Testing for an Association Between Two Categorical Variables	All				
		Slides: 7.2 Testing for an Association Between Two Categorical Variables	All			Test 3 Retake – Mon Nov 23 – Sun Nov 29 (optional; higher score counts)	None
		StatKey – Theoretical Distributions (χ^2)	MO3				